

Quantum L^2 -KKL for Observables with Commuting Pauli Support

A complementary substantial partial result

Automated Mathematics Run `fa_banach_001`, lane 4

27 August 2026

Status. `candidate_partial_likely_valid`. The full quantum L^2 -KKL conjecture remains open. This packet proves it for every balanced quantum Boolean observable whose Pauli-Fourier support is contained in an abelian Pauli subgroup.

1 Source problem

Blecher, Gao, and Xu, *Geometric influences on quantum Boolean cubes*, arXiv:2409.00224, Section 7.1 on PDF page 35, state that the original Montanaro–Osborne quantum KKL conjecture for L^2 -influence remains open.

which gives that $\|A - C_j\|_2^2 \leq 6K\varepsilon$. By letting $K_1 = 6K$, we finish the proof. $\square$

7. SUMMARY AND DISCUSSIONS

We end this paper with some further discussions.

7.1. Quantum L^2 -KKL theorem. Follow the work [7], we use the method of random restriction to obtain a dimension free bound for quantum L^1 -influence, which implies the quantum KKL-theorem for L^1 influence proved in [27]. Same as [27], our method also works with L^p -influence with $1 \leq p < 2$ but with a blow-up constant $C_p \rightarrow \infty$ as $p \rightarrow 2$ (see Remark 3.13). For higher order quantum L^1 -influence, we obtain the KKL type theorem and Talagrand inequality with constant independent of order k . This extends the classical case work [26] to the quantum setting. From this regard, the original conjecture of quantum KKL theorem (as well as high order case) for L^2 -influence from [21] remains open.

Figure 1: The open-problem passage in Section 7.1 of the source paper.

In its balanced form, the conjecture asks whether every self-adjoint unitary $A \in M_2(\mathbb{C})^{\otimes n}$ with normalized trace $\tau(A) = 0$ satisfies

$$\max_{1 \leq j \leq n} \text{Inf}_j^2(A) \geq c \frac{\log n}{n}$$

for an absolute $c > 0$. Here d_j removes the Pauli terms that are the identity on qubit j , and $\text{Inf}_j^2(A) = \|d_j A\|_2^2$.

2 Statement and proof intuition

Write $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$ for the Hermitian Pauli basis, with irrelevant signs handled separately. A Pauli-supported observable is called *commuting* here if all Pauli strings with nonzero Fourier coefficient lie in a common abelian Pauli subgroup.

The apparent obstacle is that a global Clifford diagonalizing that subgroup need not preserve physical qubits, so classical KKL cannot simply be quoted in the original coordinates. The repair is linear-algebraic. A rank- m commuting Pauli subgroup is an m -dimensional isotropic subspace of $\mathbb{F}_2^{2n}$. Its $2n$ scalar X/Z coordinate projections span the dual space, so m of them form a dual basis. Relabel the Fourier characters by that basis. Classical KKL then finds a large influence in one chosen scalar projection, and that event is contained in “the Pauli is nonidentity on the associated physical qubit.” Thus the classical influence is bounded above by a genuine quantum influence.

Theorem 1 (Commuting-Pauli quantum KKL). *There is an absolute constant $c > 0$ with the following property. Let $A = A^*$, $A^2 = I$, and $\tau(A) = 0$ on n qubits. If the Pauli-Fourier support of A is contained in an abelian Pauli subgroup, then*

$$\max_{1 \leq j \leq n} \text{Inf}_j^2(A) \geq c \frac{\log(n+1)}{n}.$$

Consequently the Montanaro–Osborne conjecture holds for this entire class.

3 Proof

Choose independent commuting Hermitian Pauli strings $P_1, \dots, P_m$ generating the subgroup spanned by the Fourier support, and put

$$P_\xi = P_1^{\xi_1} \cdots P_m^{\xi_m}, \quad \xi \in \mathbb{F}_2^m.$$

The generators may be chosen independent modulo scalar phases, so the P_ξ are distinct Pauli-basis elements and $m \leq n$. The latter bound is also the standard fact that an isotropic subspace of the symplectic space $\mathbb{F}_2^{2n}$ has dimension at most n .

Expand

$$A = \sum_{\xi \in \mathbb{F}_2^m} a_\xi P_\xi.$$

The commuting generators have all 2^m joint sign patterns, each with equal multiplicity 2^{n-m} . Indeed, the joint spectral projector for $x \in \{\pm 1\}^m$ is

$$E_x = 2^{-m} \prod_{r=1}^m (I + x_r P_r),$$

and $\text{tr}(E_x) = 2^{n-m}$ because every nonidentity Pauli string is traceless. Hence A is the functional calculus of

$$f(x) = \sum_{\xi \in \mathbb{F}_2^m} a_\xi x^\xi.$$

The conditions $A^2 = I$ and $\tau(A) = 0$ say respectively that $f : \{\pm 1\}^m \rightarrow \{\pm 1\}$ is Boolean and balanced. Moreover a_ξ are exactly its Walsh–Fourier coefficients.

Represent an element P_ξ modulo phase by its binary symplectic vector

$$\rho(\xi) = (x_1(\xi), \dots, x_n(\xi), z_1(\xi), \dots, z_n(\xi)) \in \mathbb{F}_2^{2n}.$$

The linear map $\rho : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^{2n}$ is injective. Therefore its $2n$ scalar coordinate functionals span $(\mathbb{F}_2^m)^*$. Select a basis $\ell_1, \dots, \ell_m$ from these functionals. For each r , let $j(r)$ be the physical qubit from whose X - or Z -coordinate ℓ_r was selected.

The map

$$T : \mathbb{F}_2^m \longrightarrow \mathbb{F}_2^m, \quad T\xi = (\ell_1(\xi), \dots, \ell_m(\xi))$$

is invertible. An invertible linear change of Boolean variables induces the dual relabeling of Walsh characters. Thus there is a balanced Boolean function g on m bits such that

$$\widehat{g}(T\xi) = a_\xi \quad (\xi \in \mathbb{F}_2^m).$$

For its r -th classical influence,

$$\begin{aligned} \text{Inf}_r(g) &= \sum_{\eta: \eta_r=1} |\widehat{g}(\eta)|^2 = \sum_{\xi: \ell_r(\xi)=1} |a_\xi|^2 \\ &\leq \sum_{\xi: (x_{j(r)}(\xi), z_{j(r)}(\xi)) \neq (0,0)} |a_\xi|^2 = \text{Inf}_{j(r)}^2(A). \end{aligned}$$

The inequality holds because a nonzero selected scalar coordinate forces the Pauli string to be nonidentity on that qubit; the last equality is Pauli orthogonality.

The classical KKL theorem, in a form valid for all $m \geq 1$, gives an absolute $c_0 > 0$ such that

$$\max_{1 \leq r \leq m} \text{Inf}_r(g) \geq c_0 \frac{\log(m+1)}{m} \text{Var}(g) = c_0 \frac{\log(m+1)}{m}.$$

Finally, $t \mapsto \log(t+1)/t$ decreases on $(0, \infty)$ and $m \leq n$. Combining the last two displays proves Theorem 1 with $c = c_0$. $\square$

4 Verification and limitations

The proof was checked in two independent ways. First, the only dimension count is the injective-map identity $\text{rank}(\rho) = \text{rank}(\rho^*) = m$, which guarantees the selected dual basis. Second, the accompanying script `code/verify_commuting_transfer.py` generates small random isotropic Pauli subspaces and balanced Boolean functions and verifies the exact transfer inequality $\text{Inf}_r(g) \leq \text{Inf}_{j(r)}^2(A)$ for every selected coordinate. The script is a consistency check, not part of the proof.

This result is complementary to the existing run packet for arXiv:2209.07279 on anticommuting Pauli support, which obtains the stronger order $n^{-1/2}$. Neither theorem handles a general unitary Pauli expansion, where commuting products from different parts of the support can cancel in the equations imposed by $A^2 = I$.

5 Novelty check and review recommendation

A bounded search through 27 August 2026 used the source arXiv id and the exact phrases “commuting Pauli quantum KKL”, “abelian Pauli influence KKL”, “Pauli support KKL”, and “stabilizer quantum KKL”. It recovered the known L^p , $p < 2$, quantum KKL papers and this run’s anticommuting partial result, but no theorem matching the commuting-support reduction above. Novelty confidence is moderate rather than definitive: the proof is short enough that it may be folklore through stabilizer diagonalization.

Human-review recommendation. Review as a likely-valid substantial partial result. The highest-value checks are: (i) that independent commuting Pauli generators realize all joint sign patterns; (ii) that the $2n$ local scalar projections span the dual of their isotropic subspace; and (iii) the direction of the influence comparison after the Fourier-coordinate change.

References

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- [4] C. Rouzé, M. Wirth, and H. Zhang, *Quantum Talagrand, KKL and Friedgut's theorems and the learnability of quantum Boolean functions*, Comm. Math. Phys. 405 (2024), Article 95; arXiv:2209.07279.